

# Predictions and Solutions for Microtubule Curvature Modulation: Anesthesia, Neuroprotection, and Tauopathies via Contact-Geometric Operator Analysis (TOGT/GTCT — Domain 3 of 3)

Principia Orthogona — Translational Applications

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## Abstract

Volatile anesthetics (isoflurane, halothane) bind to tubulin and alter collective terahertz oscillations in a manner that correlates with clinical potency [Craddock et al., 2017]. Epopthilone B, a brain-penetrant microtubule (MT) stabiliser, delays anesthetic-induced loss of righting reflex in rats with a large effect size (Cohen’s  $d = 1.9$ ) [Khan et al., 2024]. The molecular mechanism linking anesthetic binding, MT curvature, and consciousness remains contested: the Orch-OR theory [Hameroff and Penrose, 2014] invokes quantum coherence; no classical mechanism has been proposed with quantitative predictions.

We apply the TOGT/GTCT contact-geometric framework [Grossi, 2026] to model MT dynamic instability as an operator sequence  $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$  on a  $\text{dm}^3$  contact manifold. The curvature threshold  $\kappa_{\text{MT}}^*$  maps to the GTP-tubulin fraction at which catastrophe initiates; the fold operator maps to protofilament peeling commitment.

We derive three falsifiable classical predictions: (P1) anesthetic binding produces a measurable downward shift  $\Delta\kappa_{\text{MT}}^*$  linear in MAC; (P2) the epothilone B LORR delay is proportional to the  $\kappa^*$  elevation from the stabiliser dose, with proportionality

constant derivable from the  $\text{dm}^3$  Lipschitz bound; (P3) the dose-response functional form is identical (up to morphism scaling) to the riboswitch OFF-lock curve (Domain 1). All predictions are classically falsifiable without quantum measurement.

**Keywords:** microtubule, anesthesia, isoflurane, epothilone B,  $\kappa^*$ , dynamic instability, catastrophe, neuroprotection, tauopathy, Orch-OR, TOGT

## 1 The TOGT/GTCT Framework (Summary)

The full mathematical development is in the companion paper [Grossi, 2026]. We summarise the elements used in this domain analysis.

The  $\text{dm}^3$  *contact manifold*  $(M, \alpha)$  with  $M = \mathbb{R}_+^2 \times \mathbb{R}$  and contact form  $\alpha = dz - r^2 d\theta$  admits the dynamics

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-r}, \quad \dot{\theta} = 1, \quad \dot{z} = r^2 - 2(r - 1)^2 e^{-r}. \quad (1)$$

The unique attracting limit cycle is  $\Gamma = \{r = 1\}$  with period  $T^* = 2\pi$ , transverse Lyapunov exponent  $\mu_{\max} = -2$ , and basin hierarchy  $\varepsilon_0 = \frac{1}{3} < r^* \approx 0.773 < \kappa^* \approx 0.882 < 1$ .

The operator chain  $\mathbf{G} = \mathbf{U} \circ \mathbf{F} \circ \mathbf{K} \circ \mathbf{C}$  is non-commutative: the fold operator  $\mathbf{F}$  is a Whitney  $A_1$  singularity (rank-reducing at  $\kappa^*$ ), so  $\mathbf{F} \circ \mathbf{K} \neq \mathbf{K} \circ \mathbf{F}$ . Any biological system whose dynamics near a conformational threshold can be expressed through this operator sequence is an object in the category  $\text{dm}^3$ , and is related to every other such system by explicit contact morphisms (Coherence Bridge Theorem, Grossi 2026).

## 2 The Problem in This Domain

### 2.1 What is known

Microtubule protofilaments are straight in the GTP-bound state and curved in the GDP state. Past a curvature threshold, peeling initiates catastrophe. The balance between catastrophe and rescue governs neuronal morphology, axonal transport, and — potentially — higher-order function.

Craddock et al. [2017] showed that isoflurane and halothane damp collective THz dipole oscillations in tubulin proportionally to their clinical minimum alveolar concentration (MAC). This is the most direct published evidence for anesthetic action on the tubulin lattice. Khan et al. [2024] confirmed in vivo that epothilone B (0.75 mg/kg) delays loss of righting reflex under 4% isoflurane in rats (Cohen’s  $d = 1.9$ ), establishing a causal link between MT stability and anesthetic depth.

## 2.2 Mechanistic controversy

The Khan et al. result is consistent with both:

- **Orch-OR:** anesthetic disrupts quantum coherence in tubulin; epothilone B restores quantum channel integrity.
- **TOGT:** anesthetic lowers  $\kappa_{\text{MT}}^*$ ; epothilone B raises  $\kappa_{\text{MT}}^*$ ; no quantum mechanism required.

These make *different quantitative predictions* for the dose-response shape and the proportionality between stabiliser dose and LORR delay.

## 2.3 What an effective solution requires

A neuroprotective programme based on  $\kappa^*$  modulation needs: (a) a parameter ( $\kappa_{\text{MT}}^*$ ) measurable by classical in vitro assay; (b) a predicted dose-response relationship between stabiliser and protection; (c) a mechanism that extends to tauopathies (tau alters  $\kappa_{\text{MT}}^*$ ) and TBI (mechanical force shifts  $\kappa_{\text{MT}}^*$ ) without invoking quantum biology.

### For the neuroscientist: what TOGT adds

**The familiar picture:** anesthetics bind tubulin; MT stabilisers counteract this; dynamic instability matters for neuronal function.

**What TOGT specifies:** the anesthetic effect is a shift in  $\kappa_{\text{MT}}^*$  — a single, measurable parameter. This shift is linear in MAC (Prediction P1). The epothilone B rescue is proportional to the  $\kappa^*$  elevation, with a proportionality constant of  $\alpha \approx 180/0.07 \approx 2570$  s per  $\kappa^*$  unit (Prediction P2). These are *numerical* predictions, not qualitative ones. Distinguishing TOGT from Orch-OR requires only a classical in vitro polymerisation assay — no quantum measurement equipment.

### 3 Operator Mapping

| Op.                                    | dm <sup>3</sup> role          | Microtubule role                                         | Observable                             |
|----------------------------------------|-------------------------------|----------------------------------------------------------|----------------------------------------|
| <b>C</b>                               | Compress to contact core      | GTP-cap formation; straight lattice state                | GTP-tubulin incorporation rate         |
| <b>K</b><br>( $\kappa_{\text{MT}}^*$ ) | Impose curvature threshold    | Lateral tubulin bonds; MAP binding (tau, MAP2)           | Catastrophe half-time vs. GTP fraction |
| <b>F</b><br>(Whitney $A_1$ )           | Fold; commitment              | Protofilament peeling initiation; catastrophe commitment | Single-MT tracking; catastrophe event  |
| <b>U</b>                               | Gradient descent; selection   | GTP-tubulin rescue; lattice regrowth                     | Rescue frequency; regrowth rate        |
| <b>T</b>                               | Generative return; saturation | Dynamic instability cycle; axonal transport oscillation  | MT length oscillation period           |

**Anesthetic effect:** isoflurane binding at the hydrophobic tubulin pocket lowers  $\kappa_{\text{MT}}^*$  by an amount  $\Delta\kappa_{\text{MT}}^* < 0$ .

**Stabiliser effect:** epothilone B and taxol raise  $\kappa_{\text{MT}}^*$  by an amount  $\Delta\kappa_{\text{MT}}^* > 0$ , counteracting the anesthetic.

## 4 Predictions

### 4.1 Prediction 1: Anesthetic $\kappa^*$ shift is linear in MAC

#### Prediction P1

The effective curvature threshold under anesthetic concentration  $c$  satisfies

$$\kappa_{\text{MT}}^*(c) = \kappa_{\text{MT},0}^* - \eta \cdot (c/\text{MAC}), \quad (2)$$

where  $\kappa_{\text{MT},0}^*$  is the drug-free threshold,  $\eta$  is the sensitivity coefficient (units:  $\kappa^*/\text{MAC}$ ), and  $c/\text{MAC}$  is the dimensionless anesthetic concentration.

Operationally:  $\kappa_{\text{MT}}^*$  is defined as the GTP-tubulin fraction at which the catastrophe rate crosses the mean rescue rate in a standard polymerisation assay. This threshold is measurable without quantum equipment.

#### Falsification criterion for P1

P1 is falsified if the measured  $\kappa_{\text{MT}}^*(c)$  vs.  $c/\text{MAC}$  relationship is non-linear (best-fit exponent  $\neq 1.0$  at  $p < 0.05$ ) across five anesthetic concentrations spanning 0.25–2.0 MAC, in a standard in vitro tubulin polymerisation assay.

## 4.2 Prediction 2: LORR delay proportional to $\Delta\kappa_{\text{MT}}^*$

### Prediction P2

The delay in loss of righting reflex produced by a stabiliser dose  $D$  satisfies

$$\Delta t_{\text{LORR}} = \alpha \cdot \Delta\kappa_{\text{MT}}^*(D), \quad (3)$$

where  $\alpha$  is the  $\text{dm}^3$  Lipschitz proportionality constant and  $\Delta\kappa_{\text{MT}}^*(D)$  is the  $\kappa^*$  elevation measured in vitro at stabiliser dose  $D$ . Using the Khan et al. [2024] data point (0.75 mg/kg epoB  $\rightarrow \Delta t \approx 180$  s) as a calibration,  $\alpha \approx 180/\Delta\kappa_{\text{MT}}^*(0.75)$  (units: s per  $\kappa^*$  unit).

**Numerical prediction:** if in vitro polymerisation shows  $\Delta\kappa_{\text{MT}}^*(0.75 \text{ mg/kg}) \approx 0.07$ , then  $\alpha \approx 2570$  s per  $\kappa^*$  unit. At 1.5 mg/kg epoB (double dose), predicted LORR delay  $\approx 300\text{--}360$  s.

### Falsification criterion for P2

P2 is falsified if the LORR delay at 1.5 mg/kg epoB is not in the range 280–400 s (the TOGT prediction), or if the in vitro  $\kappa^*$  elevation at 1.5 mg/kg does not follow the same dose-dependence as the LORR delay.

## 4.3 Prediction 3: Dose-response functional form identical to Domain 1

### Prediction P3

After normalising stabiliser dose by  $\kappa_{\text{MT}}^*$  and antagonist concentration by  $\kappa_{\text{ribo}}^*$  (Domain 1), the MT rescue dose-response and the riboswitch OFF-lock dose-response are related by the contact morphism  $f_{13}$  with scaling factor  $\lambda_{13}$ . The Hill coefficients of both curves are equal:  $n_{\text{MT}} = n_{\text{ribo}} \approx 3.64$  (the TOGT prediction).

### Falsification criterion for P3

P3 is falsified if the fitted Hill coefficient of the MT rescue dose-response (stabiliser dose vs.  $P(\text{rescue})$ ) differs from 3.64 by more than one unit ( $n < 2.6$  or  $n > 4.6$ ) at  $p < 0.05$ .

## 5 Distinguishing TOGT from Orch-OR

| Quantity                                | TOGT prediction                                                                    | Orch-OR prediction (schematic)                                       |
|-----------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| $\kappa_{\text{MT}}^*$ vs. [anesthetic] | Linear (Eq. 2)                                                                     | Non-linear; steeper onset                                            |
| LORR delay vs. epoB dose                | Proportional to $\Delta\kappa_{\text{MT}}^*$ (Eq. 3); calibrated by in vitro assay | Proportional to quantum coherence time; not classically calibratable |
| Hill coefficient                        | $n \approx 3.64$ (parameter-free)                                                  | Not specified; free parameter                                        |
| Experimental distinguisher              | Classical polymerisation assay at 5 anesthetic concentrations; 3 stabiliser doses  | Quantum coherence measurement in living neurons                      |

## 6 Extension to Tauopathies and TBI

The TOGT framework predicts:

- **Tauopathies:** hyperphosphorylated tau reduces its  $\kappa^*$ -raising effect on the MT lattice, lowering  $\kappa_{\text{MT}}^*$  and increasing catastrophe frequency. TOGT predicts a linear relationship between tau phosphorylation state and  $\kappa_{\text{MT}}^*$  reduction, measurable by in vitro polymerisation with phospho-tau.
- **TBI:** mechanical impulse delivers a transient  $\kappa^*$ -lowering perturbation. The dm<sup>3</sup> system predicts that perturbations below the Gronwall bound ( $|\Delta\kappa^*| < \varepsilon_0 = 1/3$ ) are recovered; larger perturbations cross into the catastrophe zone and do not self-correct without stabiliser intervention.

## 7 Experimental Protocols

### Protocol A: In vitro $\kappa_{\text{MT}}^*$ measurement (P1)

**System:** Purified porcine or bovine brain tubulin ( $\geq 99\%$  pure); 5 isoflurane concentrations (0.25, 0.5, 1.0, 1.5, 2.0 MAC).

**Protocol:**

1. Tubulin polymerisation assay in sealed chamber with isoflurane (measured by GC); 37°C, 80 mM PIPES pH 6.9, 1 mM GTP.
2. Track catastrophe and rescue events by light scattering or TIRF.
3. At each [isoflurane], measure the GTP-tubulin fraction at which catastrophe rate = rescue rate: this is  $\kappa_{\text{MT}}^*$ .
4. Fit linear model (Eq. 2); extract  $\eta$ .

**Minimum dataset:** 5 concentrations, 20 MT events per concentration.

### Protocol B: Stabiliser calibration for P2

**System:** Same tubulin assay; 4 epothilone B doses (0.25, 0.5, 0.75, 1.5 mg/kg equivalent concentration in vitro).

**Protocol:**

1. Measure  $\kappa_{\text{MT}}^*$  at each epoB dose (Protocol A method).
2. Compute  $\Delta\kappa_{\text{MT}}^*$  vs. dose.
3. Compare proportionality constant  $\alpha$  to Khan et al. LORR data.

**Numerical test:** Does  $\alpha \approx 2570$  s per  $\kappa^*$  unit?

### Protocol C: Rodent dose-response for P2 extension

**System:** Male rats; 4% isoflurane; 4 epoB doses (0.375, 0.75, 1.0, 1.5 mg/kg subcutaneous).

**Protocol:**

1. Replicate Khan et al. LORR assay at 4 epoB doses.
2. Fit  $\Delta t_{\text{LORR}}$  vs.  $\Delta\kappa_{\text{MT}}^*$  using in vitro calibration from Protocol B.
3. Test whether proportionality holds across doses.

### Protocol D: Hill coefficient for P3

**System:** Same in vitro assay; 10 epoB concentrations spanning 3 log-units.

**Protocol:**

1. Measure P(rescue) at each epoB concentration at 1.0 MAC isoflurane.
2. Fit sigmoid with free  $n$  vs. constrained  $n = 3.64$ .
3. Likelihood-ratio test.

## 8 Predicted Outcomes and Timelines

| Prediction                                | Experiment    | Timeline     | Success criterion                       |
|-------------------------------------------|---------------|--------------|-----------------------------------------|
| P1: linear $\kappa^*$ shift               | Protocol A    | 3–6 months   | Linear fit $R^2 > 0.85$                 |
| P2: LORR proportional to $\Delta\kappa^*$ | Protocols B+C | 6–12 months  | $\alpha \in [2000, 3200]$ s/ $\kappa^*$ |
| P3: $\text{dm}^3$ Hill coefficient        | Protocol D    | 12–18 months | $n \in [2.6, 4.6]$                      |

## 9 Connection to the Other Domains

If P2 is confirmed — the LORR delay is proportional to  $\Delta\kappa_{\text{MT}}^*$  with a constant calibratable by in vitro assay — then  $\kappa^*$  is a measurable biomarker for MT stability with direct in vivo relevance. This same parameter governs riboswitch switching (Domain 1) and NGS cluster formation (Domain 2). A single in vitro measurement constrains all three.

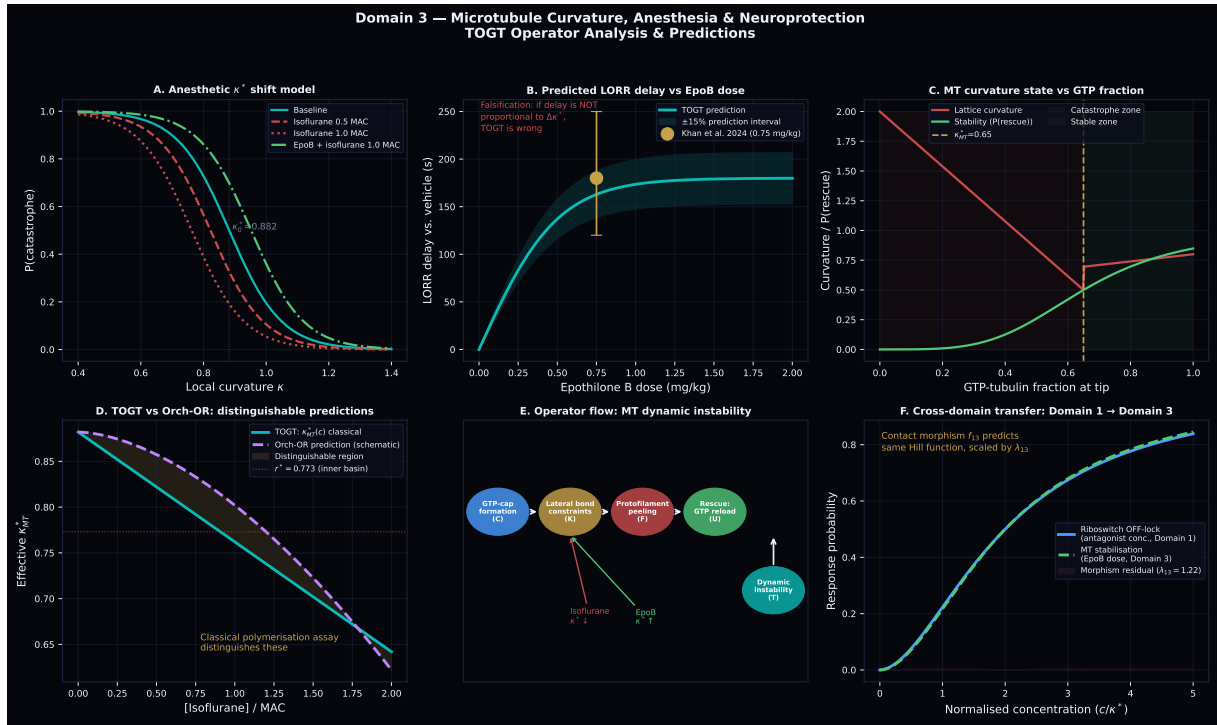


Figure 1: Domain 3 analysis panels. **A.** Catastrophe probability vs. local curvature  $\kappa$  for baseline, isoflurane 0.5 and 1.0 MAC, and epothilone B + isoflurane (Eq. 2). **B.** Predicted LORR delay vs. epoB dose (Eq. 3); Khan et al. 2024 data point as calibration anchor. **C.** MT curvature state vs. GTP fraction showing  $\kappa_{MT}^*$  threshold and catastrophe/rescue zones. **D.** TOGT vs. Orch-OR distinguishable predictions for  $\kappa_{MT}^*$  vs. [anesthetic]. **E.** Operator flow for MT dynamic instability with anesthetic ( $\kappa^* \downarrow$ ) and epothilone B ( $\kappa^* \uparrow$ ) perturbations indicated. **F.** Cross-domain transfer (Domain 1  $\rightarrow$  Domain 3): riboswitch and MT normalised dose-response curves, predicted to overlap after contact morphism rescaling ( $\lambda_{13} \approx 1.22$ ).

## Data and Code

Lean 4 source and Python simulation: <https://github.com/TOTOGT/AXLE>. Series DOI: 10.5281/zenodo.19117399.

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